



University of Asia Pacific

Department of Computer Science and Engineering

CSE 316: Microprocessors and Microcontrollers Lab CEP Mapping

Submitted by:

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CEP mapping

1. Knowledge Profile (K's) Mapping

K's Attributes	How K's Are Addressed Through Our Project	CLO	PLO
K1 – Engineering Fundamentals	Skills in ESP32 programming, sensor integration, relay and servo actuation, and IoT cloud communication.	CLO1, CLO2	PLO1, PLO2
K2 – Specialist Knowledge	Knowledge of environmental sensors (DHT11, soil moisture, HC-SR04, rain sensor) and Blynk IoT platform for real-time monitoring.	CLO2, CLO3	PLO1, PLO2
K3 – Engineering Design	Designing modular system architecture integrating sensing, actuation, and cloud communication units.	CLO3, CLO4	PLO3, PLO5
K4 – Engineering Practice	Implemented IoT-based plant hydration system using ESP32, sensors, relay, servo, and water pump following circuit and coding best practices.	CLO4, CLO5	PLO4, PLO5, PLO6
K5 – Comprehension	Ensured system operates reliably under varying environmental conditions with automated and manual control options.	CLO5	PLO7

2. Complex Engineering Problems (P's) Mapping

P's Attributes	How P's Are Addressed Through Our Project	CLO	PLO
P1 – Depth of Knowledge Required	Required understanding of multiple sensors, actuator control, ESP32 programming, and IoT communication protocols.	CLO1, CLO2, CLO4	PLO1, PLO2
P2 – Depth of Analysis Required	Analyzing environmental data (soil moisture, temperature,	CLO2, CLO3	PLO1, PLO2,

P's Attributes	How P's Are Addressed Through Our Project	CLO	PLO
	humidity, rainfall, water level) and deciding when to activate pump and shed.		PLO6
P3 – Interdependence	Integration of sensors, relay, servo, ESP32, and Blynk platform to achieve synchronized and autonomous operation.	CLO1, CLO3, CLO5	PLO1, PLO3, PLO7

3. Complex Engineering Activities (A's) Mapping

A's Attributes	How A's Are Addressed Through Our Project	CLO	PLO
A1 – Range of Resources Used	ESP32 Devkit, DHT11, HC-SR04, capacitive soil sensor, rain sensor, relay, servo motor, water pump, and Blynk IoT dashboard.	CLO1, CLO4	PLO1, PLO6
A2 – Level of Interaction	Integration of multiple sensors, actuators, ESP32 firmware, and cloud communication with real-time user monitoring and manual overrides.	CLO2, CLO3	PLO3, PLO7
A3 – Innovation	Combines automated plant hydration, rain-shed protection, and IoT-based remote monitoring in a single low-cost system.	CLO3, CLO5	PLO4, PLO5, PLO7
A4 – Consequences for Society & Environment	Promotes efficient water usage, reduces manual labor in home gardening or small-scale agriculture, and provides a scalable smart agriculture solution.	CLO5, CLO6	PLO7